NAI
Najah AI SOCIETY

Data Analysis and Machine Learning

NAI Course By Mr.Mosab Adwan

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INTRODUCTION

We were given a csv file that contains data for the bank .First I took a look at the type of data we are dealing with .The data is structured .Data type categorical data with numerical data .

	b	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	deposit
	dmin.	married	secondary	no	2343	yes	no	unknown	5	may	1042	1	-1	0	unknown	yes
	dmin.	married	secondary	no	45	no	no	unknown	5	may	1467	1	-1	0	unknown	yes
	chniciar	married	secondary	no	1270	yes	no	unknown	5	may	1389	1	-1	0	unknown	yes
55	services	married	secondary	no	2476	yes	no	unknown	5	may	579	1	-1	0	unknown	yes
54	admin.	married	tertiary	no	184	no	no	unknown	5	may	673	2	-1	0	unknown	yes
42	managem	single	tertiary	no	0	yes	yes	unknown	5	may	562	2	-1	0	unknown	yes
56	managem	married	tertiary	no	830	yes	yes	unknown	6	may	1201	1	-1	0	unknown	yes
60	retired	divorced	secondary	no	545	yes	no	unknown	6	may	1030	1	-1	0	unknown	yes
37	technicial	married	secondary	no	1	yes	no	unknown	6	may	608	1	-1	0	unknown	yes
28	services	single	secondary	no	5090	yes	no	unknown	6	may	1297	3	-1	0	unknown	yes
38	admin.	single	secondary	no	100	yes	no	unknown	7	may	786	1	-1	0	unknown	yes
30	blue-colla	married	secondary	no	309	yes	no	unknown	7	may	1574	2	-1	0	unknown	yes
29	managem	married	tertiary	no	199	yes	yes	unknown	7	may	1689	4	-1	0	unknown	yes
46	blue-colla	single	tertiary	no	460	yes	no	unknown	7	may	1102	2	-1	0	unknown	yes
31	technicial	single	tertiary	no	703	yes	no	unknown	8	may	943	2	-1	0	unknown	yes
35	managem	divorced	tertiary	no	3837	yes	no	unknown	8	may	1084	1	-1	0	unknown	yes
32	blue-colla	single	primary	no	611	yes	no	unknown	8	may	541	3	-1	0	unknown	yes
49	services	married	secondary	no	-8	yes	no	unknown	8	may	1119	1	-1	0	unknown	yes
41	admin.	married	secondary	no	55	yes	no	unknown	8	may	1120	2	-1	0	unknown	yes
49	admin.	divorced	secondary	no	168	yes	yes	unknown	8	may	513	1	-1	0	unknown	yes

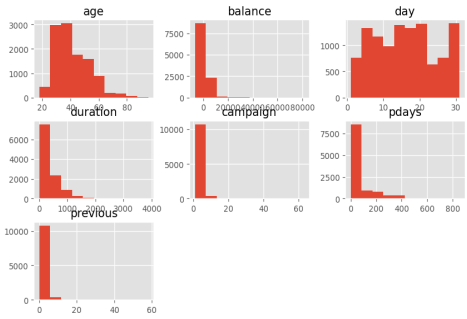
Tools Used

1. Jupter Note book
2. Pandas Library
3. Matplotlib.pyplot
4. sklearn import datasets
5. seaborn

Procedure And Analysis

At First I Looked into the data:

- Statistical info
- Data dimensions,columns and rows ,etc..
- Categorical values and features of the data



Encoding

We Encoded any categorical values so we use it in upcoming tasks

We Also Took The chance to remove any outliers before classifying the data

```
pdays outliers: [166  91 147 ... 271 172 172]
previous outliers: [4 3 4 ... 5 4 5]
```

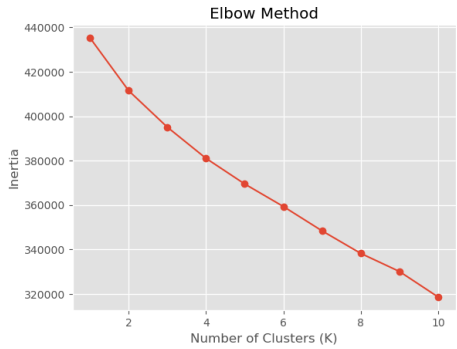
K-Means (classifying)

Data needs to be Grouped before the machine gets to use our analyze

Data was grouped into clusters

We measured the inertia which helps us in knowing how many clusters there are and how far are they from the center .

Elbow Visual:



Supervised Learning

Data Preparation

Before training the models, the dataset was prepared as follows:

- The target variable `poutcome` was separated from the input features.
- Categorical variables were converted into numerical format using one-hot encoding.
- The dataset was split into testing (training (80% /20%) sets using stratified sampling to preserve class distribution.

Two models were used to Analyze :

Model one :

Neural Network

A fully connected neural network was implemented using one hidden layer(mlpclassifier use).

Hidden layer: 100 neurons

Activation function: ReLU (Rectified Linear Unit)

Optimizer: Adam

Loss function: Cross-Entropy (used internally for classification)

Model Two

Random Forest (Ensemble Method)

A Random Forest classifier was used as an ensemble learning method.

Configuration

- Number of trees: 100
- Maximum depth: 10

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Comparison

Which model performed best?

Based on the experimental results, the Random Forest model outperformed the Neural Network.

Random Forest Accuracy: 91.54%

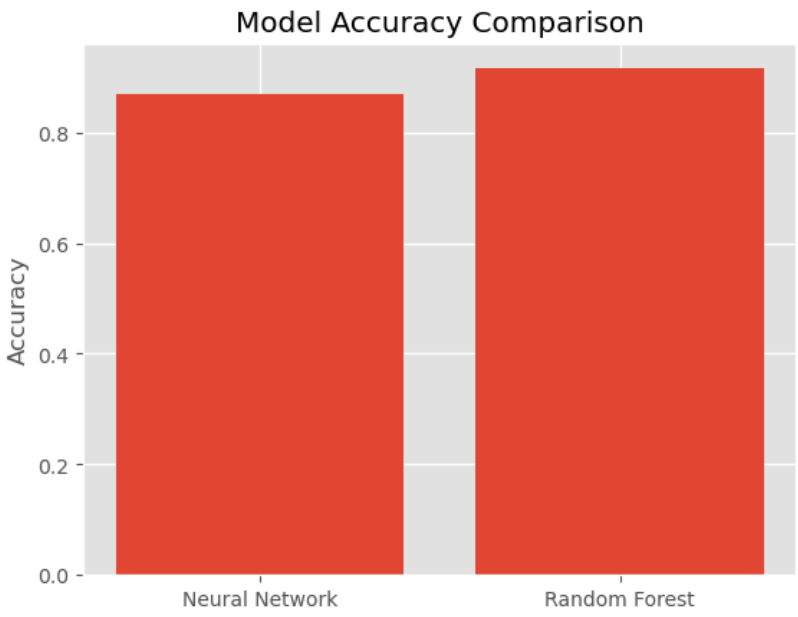
Neural Network Accuracy: 86.97%

the Random Forest showed better generalization:

RF Train: 92.91% vs Test: 91.54%

NN Train: 99.61% vs Test: 86.97%

One of the problems faced is that the Neural network model needed a lot of iterations because on small values he kept giving errors ,Needed more Iterations to train on more data and learn



Effect of Hyperparameter Tuning

Hyperparameter tuning improved model performance and stability, especially for

Random Forest.

the Neural Network remained sensitive to overfitting despite tuning

model still showed overfitting, as seen from the large gap between training and testing accuracy (problem # 2)

Experiment Example:

Experiment Setup

Neural Network

Tested parameters:

Hidden layers: 50, 100

Learning rate: 0.001

Batch size: 32

Random Forest

Tested parameters:

Number of estimators: 100, 150

Max depth: 10

Results

Before Tuning:

Neural Network Accuracy ≈ 86.97%

Random Forest Accuracy ≈ ~90% (baseline closer)

After Tuning:

Neural Network Accuracy = 86.97% (no improvement)

Random Forest Accuracy = 91.54

Matrix Analysis

The model performd well overall

But performs poorly on some classes

Reasons and analysis

The dataset likely contains different unblanced classes

Minority classes were often out of calculation

The model is biased toward the majority class

For Random Forest

Likely achieved Better classification overall

Problems Faced :

Categorical Data Handling

Models could not process text values directly so they Required encoding

Model Training Issues

Required tuning of iterations

Class Imbalance

Some classes had fewer samples, affecting model performance

Improvements

Trying better neural network models

Try Gradient Boosting (e.g., XGBoost) for better performance

The End ,Report by Saed Swidan

Thank you *Mr.Mosab Adwan* for this amazing course ,I really benefited from it